SWORD v17c Beta Release Notes

Version: v17c beta 0.0.13 **Date:** October 2026 **Base version:** SWORD v17b (March 2025, UNC)

Changelog

0.0.13 (October 2026)

- **facc rederived on all nodes and reaches.** `facc` (upstream drainage area, km^2) on all 11,112,454 nodes and 248,673 reaches is replaced by a rederivation from MERIT Hydro upstream area (UPA). Each physical section (a chain of reaches between junctions; 43,218 globally) is matched to MERIT's D8 flow-direction grid, and each node takes the maximum UPA over its own stretch of the matched route. Where that section-level D8 match was withheld, a probabilistic section matcher (distance to the river as evidence, D8 connectivity as the transition model, no previous facc as input) supplies the value. Short gaps between matched values are bridged from the nearest matched values, and the remainder is filled by a network-constrained fill: the cleaned v17b profile clipped between trusted upstream and downstream values. v17b values therefore enter only through the fill. Within every reach, node `facc` never increases upstream. Reach `facc` is the integration's junction-aware reach value; it equals the maximum of the reach's nodes on all but 49 reaches (all fill), where it is lower (median 0.6%).
- **Provenance in `facc_quality`.** Every node and reach carries `facc_quality = rederived_v1:<source>:<tier>`, with `:review` appended when the row is flagged `junction_inconsistent` or `topology_suspect` (1,406 reaches, 58,538 nodes). Sources are `engine` (section D8 match), `soft` (probabilistic matcher), `bridge` (held or interpolated from the nearest matched values) and `fill` (network-constrained fill). Tiers `A`, `A2`, `B`, `C`, `D`, `E` record which rule produced the value, not a measured accuracy (Section 3). GeoPackage, GeoParquet and DuckDB carry the string. NetCDF keeps `facc_quality` as an int32 flag variable with code `100*source + 10*tier + review` and CF `flag_values/flag_meanings`. The 0.0.12 vocabulary (`denoise_v3`, or unflagged) is retired.
- **Coverage by source.** Nodes: engine 63.3%, matcher (`soft`) 32.0%, bridge 3.4%, fill 1.3%. Reaches (majority node source): engine 62.3%, matcher 32.6%, bridge 3.4%, fill 1.7%. Reach tiers (worst node tier): A 62.3%, A2 11.6%, B 13.8%, C 4.8%, D 6.9%, E 0.5%.
- **Comparison with independent references.** Share of comparisons within 20% of the reference, for v17b / released v17c (0.0.12) / 0.0.13:
  - 6,262 HydroBASINS outlets (one SWORD reach per single-exit PFAF6 unit): 72.7% / 41.9% / 84.4%; median absolute error 3.4% / 30.9% / 0.96%.
  - 2,298 SWOT Validation Set gauges on all continents, against HydroSHEDS/RiverATLAS upstream area at the gauge: 78% / 61% / 95%.
  - 101 randomly selected USGS gauges in the conterminous US with published drainage areas, compared at the nearest node: 78% / 62% / 99%.

  HydroBASINS and HydroSHEDS share SRTM ancestry with MERIT Hydro; the USGS drainage areas do not. HydroBASINS tests outlets only.
- **Change relative to 0.0.12.** Of the 248,625 reaches with a positive 0.0.12 value, 57.2% are within 20% of it, 32.8% are more than 20% lower and 10.1% are more than 20% higher (median ratio 0.98). The decreases remove the junction-sum inflation of the 0.0.12 denoise: the largest 0.0.12 reach value was 682,449,570 km^2 on the Lena delta, more than the Earth's land area. The 48 reaches and 1,546 nodes with `facc = 0` in 0.0.12 now have positive values.
- **v17b facc no longer in the product.** The v17b values that 0.0.12 kept on 152,793 reaches are replaced. The 0.0.12 `facc` and `facc_quality` values are kept in the working-database tables `facc_backup_0012_nodes` and `facc_backup_0012_reaches`; they are not in the distributed files.
- **What did not change.** Topology, IDs, counts, geometry and every other variable are identical to 0.0.12; a column-by-column comparison against the published 0.0.12 GeoParquet and NetCDF files finds differences only in `facc` and `facc_quality`. The routing and mainstem attributes (`best_headwater`, `best_outlet`, `main_path_id`, `is_mainstem`, `rch_id_up_main`, `rch_id_dn_main`, and the distances that follow them: `hydro_dist_out`, `hydro_dist_hw`, `pathlen_hw`, `pathlen_out`) were computed in 0.0.12 with the 0.0.12 `facc` as one term of the routing score (`0.17*log1p(facc)`, about 6% of the score) and were not recomputed. `area_fits` and `discharge_models` remain v17b copies.
- **facc is no longer adjusted to conserve.** 0.0.12 enforced junction conservation and downstream monotonicity by raising values, which is what produced the inflation. The 0.0.13 values are read from MERIT Hydro. The integration applies a running maximum within sections, but it does not sum upstream values at junctions and does not raise a reach to an upstream tail when the two independently matched sections disagree. The facc lint counts therefore change by design: T003 (decrease of more than 5% across a single-downstream link) 0 → 1,852;

F006 (junction conservation, severity error) 0 → 4,537 of 18,318 junctions; F001 (facc/width ratio anomaly) 601 → 140; F002 (facc jump ratio) 35 → 788; F010 (junction raise then drop) 0 → 27; F011 (link monotonicity) 646 → 352. A full lint run with `--fail-on-error` now fails on F006 only.

- **Known limitations.**
  - Sections are matched independently, so the two sides of a junction can disagree. 1,378 reaches (57,274 nodes) whose head is below 0.8 times the largest upstream tail, or below a single-downstream link that drops by more than 20%, are flagged `junction_inconsistent`; 88 reaches where a width guard refused a raise are flagged `topology_suspect`. Both carry `:review` (1,406 reaches in total).
  - A review list from the assessment (value differs from v17b by more than 25% and the reach has a risk flag: width check not ok, monotone repair, bridge or fill source, or tier C/D/E) covers 30,674 reaches (12.3%). It is not encoded in this release. At the 61 gauges inside the list, the 0.0.13 value is within 20% of the reference more often (70%) than v17b (15%) or the fill (44%).
  - The width plausibility check is fitted only for ordinary river, canal and dam reaches. Lake, tidal, ghost and other reaches, plus 888 river reaches without a usable width (66,449 reaches in total), are not checked, so wrong matcher values there are not caught.
  - Wrong SWORD topology edges carry a trusted value into the wrong river. At the Loire → Vilaine link, the guard stops the Loire value at the first canal reach (23221000021, 474 km^2), but dam reach 23221000054 and river reaches 23221000041 and 23221000031 still carry the Loire's 111,941 km^2. This needs a topology edit, not a facc rule.
  - Amazon reach 62253000041 (fill, tier C), where two braided channels rejoin, carries 9,026,821 km^2: the sum of its two upstream channels (about 4.51 million km^2 each), twice the value of the reaches above and below it.
  - 194 sections where an interior matcher spike is carried to the section tail, and 28 lake-tail latches, are listed in the assessment diagnostics and not corrected.
- **Exports regenerated and verified.** NetCDF, GeoPackage, GeoParquet and the global GeoParquet and DuckDB files were regenerated for all six regions. `facc` and `facc_quality` match the database on every node and reach in every format. NetCDF variable dtypes are unchanged from 0.0.12 and match v17b for every shared variable. The POM release gate returns the same result as 0.0.12 on all 30 checks.
- **Working-database change besides facc: per-region node views re-bound.** The six per-region views `af_nodes`, `as_nodes`, `eu_nodes`, `na_nodes`, `oc_nodes` and `sa_nodes` (`SELECT * FROM nodes WHERE region = '<R>'`) were stale before this release: columns had been added to `nodes` after the views were created, so every query on them failed with a DuckDB binder error. The `*_reaches` and `*_centerlines` views were not affected. The six views were re-created from their own stored definition (`CREATE OR REPLACE VIEW <r>_nodes AS SELECT * FROM nodes WHERE region = '<R>'`, the same statement as in `src/sword_duckdb/schema.py`); no definition, table or value other than `facc`/`facc_quality` changed. The views are not part of the distributed files. The operations are recorded in `sword_operations`: 1002 applied the rederivation, but its post-write check stopped on the stale views; 1003 was a full rollback, which also re-bound the views, and left `facc`/`facc_quality` identical to a pre-write copy of the database on every node and reach; 1004 re-applied the rederivation and passed every post-write check.
- **Method and validation.** See FACC_FABLE_ASSESSMENT_20261001.md.
- **Next.** A second version of the rederivation that matches sections jointly at junctions (`rederived_v2`) is verified and staged for 0.0.14.

0.0.12 (May 2026)

- **Node coordinates restored to v17b for SWOT D0-D2 continuity.** The 344 reaches whose nodes were rederived in 0.0.8 and 0.0.10 now use v17b node `x`/`y`, `node_length`, `cl_id_min`, and `cl_id_max` values again. A small OC split-revert residue on reach 51111300061 was included in the same pass. Post-revert validation found zero node coordinate differences relative to v17b NetCDF across all 11,112,454 nodes.
- **Centerline-to-node assignments resynchronized.** `centerlines.node_id` was restored from v17b on affected reaches so node geometry and centerline allocation remain internally consistent.
- **Node order repaired on restored-coordinate reaches.** Projection-based ordering over the v17b-restored node coordinates updates `node_order` on 3,725 nodes across 221 of the 344 affected reaches, and updates 45 `dn_node_id`/`up_node_id` boundary pairs. The repair preserves node coordinates and recomputes the six node distance fields from the current midpoint convention across all 344 restored-coordinate reaches, including a distance-only cleanup for 289 inherited `dist_out` rows on five unchanged-order reaches. AS reach 35301100891 now encodes POM's intended downstream-to-upstream sequence (`2, 3, ..., 23, 25, 26, 24, 27, ..., 73, 75, 74, 1`) without moving node lat/lon.

- **SWOT slope observations recomputed from node WSE.** `slope_obs_*` on reaches now comes from pass-level regressions of RiverSP node `wse_sm` against the current beta12 `nodes.dist_out`, then reach-level percentiles of those pass slopes. This replaces stale reach-level signed RiverSP slopes whose sign could reflect older node ordering. Median `slope_obs_p50` coverage is 173,732 reaches; negative medians decrease from 18,805 before the correction to 9,988 after the correction. POM's canonical AS reach 35301100891 changes from `slope_obs_p50=-0.0136548684` to `+0.0142687580` m/m while retaining `slope=15.121399` m/km. Downstream users should read both `slope` and `slope_obs_*` directly from 0.0.12; no external sign flip is needed for reaches whose node order changed.
- **Static MERIT/SRTM slope semantics unchanged.** The legacy `reaches/slope` variable remains the nonnegative MERIT/SRTM slope magnitude in m/km; it is not a signed flow-direction variable. 0.0.12 does not change `reaches/slope`, `reaches/wse`, or `nodes/wse` relative to v17b. Endpoint-WSE sign checks may still warn where inherited MERIT/SRTM node WSE values are locally inconsistent with flow direction; those warnings are data-quality diagnostics, not instructions to flip `reaches/slope` or `slope_obs_*`. MERIT/SRTM WSE resampling was evaluated after POM's Test 16 feedback and deferred: the legacy SWORD workflow used UPA-filtered nearest MERIT river pixels and centerline aggregation, so a faithful correction is a science-data reconstruction task rather than a final beta patch.
- **v17c analytic fields preserved.** The revert does not undo the v17c midpoint node-distance convention, lakeflag/type reconciliation, SWOT WSE/width observation statistics, metadata restoration, or export ordering changes. No RiverObs software change or coordinate-substitution mapping table is required for D0-D2 coordinate continuity.
- **Exports regenerated and verified.** NetCDF, GeoPackage, and GeoParquet artifacts were regenerated for all six regions. Export row counts match the DuckDB source for reaches and nodes in GeoPackage/GeoParquet, and for reaches, nodes, and centerlines in NetCDF.

0.0.11 (April 2026)

- **Node ordering normalized in NetCDF export.** About 7.5% of reaches (18,553 globally) had nodes stored in descending dist_out order in all previous SWORD versions due to arbitrary GRWL centerline digitization direction. Nodes are now exported in ascending node_order (downstream-first) for all reaches. This fixes slope sign reversals in processing code that assumes the first node in the array is the downstream end. The underlying data is unchanged; only the file ordering is corrected.
- **Restored metadata variables in NetCDF export.** river_name_local, river_name_en, version, add_flag, and swot_obs_source were accidentally omitted from the export spec in 0.0.4. Restored in 0.0.11 on both reaches and nodes where applicable.
- **Node lakeflag propagated from reach.** Node lakeflag now matches parent reach lakeflag across all 11,112,454 nodes, per JPL request.

0.0.10 (April 2026)

- **Node geolocation fixed on 303 additional reaches.** `rederive_nodes` recomputes node x/y from centerline spatial partitioning for reaches where consecutive node gaps exceed 3x the reach's median spacing and 0.4 km absolute (POM test 6b criteria). By region: AS:189, SA:38, AF:29, EU:21, OC:19, NA:7. Combined with the 41 reaches fixed in 0.0.8, total rederived: 344 reaches (0.14%). **10 reaches were fixed but still trigger the scrambled-node detector** (centerline geometry issues inherited from v17b); see Known Limitations.
- **Reach 35301100891 node_order rotation and geolocation fixed.** This AS reach had node_ids rotated by one position (2, 3, ..., 75, 1 instead of 1, 2, ..., 75) since v17b. Additionally, 74 of 75 nodes were rederived from the centerline (max shift ~12.6 km on node 35301100890011) due to scrambled geolocation detected by POM test 6b. `node_order` now matches `node_id` ascending, `dn_node_id` = 0011, `up_node_id` = 0751. Node dist_out and all interpolated distance columns recalculated to match the corrected node_order.
- **Reach lakeflag and type reconciled for lake classification.** About 6,200 reaches had inconsistent lakeflag and type fields (lakeflag=1 with type=1, or lakeflag=0 with type=3). Inconsistent reaches were either orphaned (skipped by both river and lake processing) or double-counted. Resolution combined three methods: 1,015 manual reviews through the Streamlit QA app, a gradient-boosted classifier trained on those reviews (82% precision, 6% FPR, applied at high confidence only: $p>0.8$ for lake, $p<0.2$ for river), and direct corrections from HarP v1.1 lake classifications. Lakeflag and type are now 100% consistent across all 248,673 reaches. The type column now diverges from the reach ID last digit on 2,316 reaches (0.9%); the type column is authoritative.
- **Six reaches fixed for N013 closure-bug damage.** Reaches 14278900061 (AF), 31241401301, 48294000081, 45570000125, 34100005185, 42211000503 (AS) had corrupted x/y and cl_id_min/cl_id_max from the N013

closure bug (documented in 0.0.8). Nodes rederived from their own centerlines using the fixed code; node_length restored from v17b NetCDF to preserve exact sum-equals-reach_length consistency. Full sweep confirms zero reaches with node_length mismatch above 0.1% globally.
- **SWOT reach filters aligned with node filters.** `build_reach_filter_sql` now includes cross-track distance (10-60 km) and valid time_str filters, matching the node-level filters. Code change only; no DB data affected.
- **`rederive_scrambled_nodes.py` safeguards added.** After rederiving nodes, the script now automatically recalculates node dist_out (prevents N004 violations) and verifies node_length sums (catches G002 regressions).

0.0.9 (April 2026)

- **Flow correction fully reverted.** All 810 flow-corrected reaches restored to v17b topology after discovering a scoring tautology in `score_section_confidence` (`slope_from_upstream = -slope_from_downstream`). Topology diff vs v17b: 0. `v17c_flow_corrections` table is empty.
- **dist_out corruption fixed.** 1,976 reaches had BFS-corrupted `dist_out` from a January `CALCULATE_DIST_OUT` bug (wrong outlet, 93–99% error). Restored v17b values. T017 violations: 701 → 557.
- **Pipeline re-run with `skip_flow_correction=True`.** All 6 regions pass all gates.

0.0.8 (April 2026)

- **Node `dist_out` unified to midpoint convention.** All six node-level distance columns (`dist_out`, `hydro_dist_out`, `dist_out_dijkstra`, `hydro_dist_hw`, `pathlen_hw`, `pathlen_out`) now use midpoint interpolation from reach-level values. Previously `dist_out` preserved v17b endpoint values while the other five used midpoint, causing a systematic ~100 m discrepancy on single-path networks. On such networks the three outlet-distance columns are now exactly equal at every node. Breaking change from v17b convention; all POM release gate checks pass.
- **Node geolocation fixed on 41 reaches.** `rederive_nodes` recomputes node x/y from centerline spatial partitioning for 41 reaches (SA:5, EU:5, AF:8, AS:20, OC:3) where source NetCDF had contradictory CL-to-node mappings causing >2 km geographic jumps between consecutive nodes. Node lengths now use boundary-to-boundary distances (sum equals `reach_length` exactly). POM example reach 13341000591: max consecutive node jump drops from 13.9 km to 223 m.
- **`rederive_nodes` closure bug fixed.** The N013 centerline sync inside `rederive_nodes` captured a stale `reach_id` from an outer loop instead of the current plan's reach. Fixed to use `plan["reach_id"]`.

0.0.7 (April 2026)

- **Flow-corrected reach node_order and dist_out restored.** When pipeline runs skip the facc stage (e.g. when only rerunning distance and mainstem computations), `flipped_reach_ids` was empty, so `update_node_columns` never reversed `node_order` or swapped `dn_node_id`/`up_node_id` for the 810 flow-corrected reaches. `0.0.6` shipped with `node_order` matching v17b `node_id` order on those reaches (Pierre-Olivier Malaterre detected 389 inverted reaches). `0.0.7` loads the flipped reach IDs from the `v17c_flow_corrections` table when facc is skipped, so the node-order reversal and `dn_node_id`/`up_node_id` swap always run.
- **Node `dist_out` recomputed for flow-corrected reaches.** After reversing `node_order`, v17b node `dist_out` values are stale on 810 reaches (computed from the wrong flow direction). `0.0.7` recomputes them using cumulative `node_length` midpoint offsets from the corrected downstream boundary, so `dist_out` is monotonic with `node_order` and matches the new flow direction.
- **New lint check N018.** Catches the regression above automatically: flags flow-corrected reaches whose `node_order` is not reversed from v17b `node_id` order. ERROR severity, wired into the POM release gate.

0.0.6 (April 2026)

- **Canonical export ordering restored.** The `0.0.5` flat-file exports were written in DuckDB storage order through the generic maintenance exporter, which does not preserve a stable logical row order unless it is told to do so. As a result, node arrays in the distributed files could be split across multiple blocks for the same reach even though the per-row attributes stayed aligned.
- **Nodes now export in reach-contiguous logical order.** `0.0.6` writes nodes ordered by `reach_id`, then `node_order`, then `node_id` across NetCDF, GeoPackage, and Parquet. This restores a clean in-file node sequence for downstream consumers that expect each reach to occupy one contiguous node block and to run from downstream (`node_order=1`) to upstream (`node_order=n`).

- **Observed regression removed.** In Africa, the count of reaches whose NetCDF node rows were split into multiple file blocks dropped from 536 in `0.0.5` to 0 in `0.0.6`. Pierre-Olivier Malaterre's example reach `13341000591` now appears as one contiguous node block with `node_order = 1..93`.

0.0.5 (April 2026)

- **Promoted corrected bundle to a new version.** The corrected April 2 reissue is now published as `0.0.5` instead of reusing `0.0.4`, so testers do not keep hitting stale downloads or cached copies under the same name. This `0.0.5` bundle supersedes all prior `0.0.4` beta uploads on Drive.
- **Release regressions from distributed `0.0.4` artifacts fixed.** The `0.0.5` bundle corrects the specific tester-reported regressions from the prior `0.0.4` uploads: reversed node `dist_out` relative to `node_order`, stale `dn_node_id` / `up_node_id` orientation on flow-corrected reaches, missing `nodes` layers in some GeoPackages, NULL `dist_out_dijkstra` at isolated ghost coastal outlets, and inconsistent single-node node-distance convention relative to the other node-level outlet distances.
- **Release artifacts renamed and resynced.** NetCDF, GeoPackage, Parquet, release notes, and SHA256 manifests now use `0.0.5` filenames consistently in the local beta folder and Google Drive beta folder.

0.0.4 (March 2026)

- **Node propagation to 11.1M nodes.** Five v17c columns (`best_headwater`, `best_outlet`, `pathlen_hw`, `pathlen_out`, `subnetwork_id`) were NULL on all nodes. Now propagated from parent reaches.
- **Dijkstra ghost outlet fix.** Ghost reaches (type=6) with out_degree=0 no longer report `dist_out_dijkstra=0`. All sinks are used as Dijkstra sources for full coverage, with outlet values following the v17b convention (`dist_out_dijkstra = reach_length`). Real outlet counts: NA=7, SA=1, EU=2, AF=1, AS=11, OC=1.
- **Bifurcation routing fix.** At multi-successor nodes, `rch_id_dn_main` now follows the mainstem chain unconditionally (was falling back to score-based ranking, which could disagree with `is_mainstem`). V023 (pathlen_out step consistency) violations: 2 → 0.
- **`hydro_dist_hw` computed.** Mainstem distance from headwater via `rch_id_up_main` chain walk (mirror of `hydro_dist_out`). Was stale from a prior pipeline run.
- **facc monotonicity fix (T003).** 419 reaches corrected via iterative downstream propagation. T003 violations: 392 → 0.
- **Routing weights retrained on `effective_slope`.** SWOT `slope_obs_p50` (where reliable and n>=5) replaces MERIT DEM slope in routing score training. Slope share: 8% → 3%, width: 67% → 71%. CV accuracy unchanged (88.5%).
- **Node-level distance interpolation.** `hydro_dist_out`, `hydro_dist_hw`, and `dist_out_dijkstra` added to nodes table, interpolated by node position within reach (using v17b `dist_out` offset). `pathlen_hw` and `pathlen_out` changed from flat reach copies to per-node interpolated values.
- **`hydro_dist_hw` convention fix.** Reach-level values shifted from downstream-end reporting to upstream-end reporting, matching `dist_out`, `hydro_dist_out`, and `dist_out_dijkstra`. Headwater reaches now report 0 (was `reach_length`). Node-level values unchanged.
- **F006 junction conservation fix.** 2 remaining junction violations (OC 53130100215, AS 45311901585) resolved by setting downstream facc to sum of upstream facc. F006 violations: 2 → 0.
- **13 AS `main_side` reverted to v17b.** Reaches had `main_side` changed from 0 (main) to 1 (side) by an undetermined prior operation; 9 of 13 are linear reaches where side-channel classification is impossible.
- **Distance convention documented.** Variable reference now includes a convention table specifying endpoint reporting, zero-point, and ghost reach behavior for all distance variables, plus node-level interpolation formulas.
- **Variable reference updated.** 7 missing variables added, 8 type mismatches fixed, `cl_ids` shape corrected to `cl_id_min`/`cl_id_max`.
- **Node dist_out reactive recalc fix.** `CALCULATE_DIST_OUT` operations now correctly recalculate node-level `dist_out` by sorting nodes via `node_order` (geometric position) instead of `node_id`. This fixes an issue where flow-corrected reaches had incorrect node distances after recalculation (e.g., downstream-most node getting downstream reach's `dist_out`).
- **April 2 artifact reissue.** The published April 1 `0.0.4` files still had stale orientation metadata on 639 flow-corrected reaches. The reissued NetCDF, GeoPackage, and Parquet bundles recompute `node_order`, `dn_node_id`, and `up_node_id` from node `dist_out`, restore `dist_out_dijkstra = reach_length` for isolated ghost coastal outlets, and align single-node `node.dist_out` with the same midpoint convention used by the other node-level outlet distances.
- **Ghost coastal outlet dist_out_dijkstra fix.** Isolated ghost coastal outlets (type=6 sinks with upstream

neighbors but no path to real hydrologic outlets) now receive `dist_out_dijkstra = reach_length`, matching v17b `dist_out` behavior. Previously these 703 reaches in NA (similar counts in other regions) had NULL, but they should report the reach-level outlet distance to the ocean outlet point.
- **Exports regenerated (April 1 and 2, 2026).** The initial April 1 bundle captured the code fixes above. The April 2 reissue refreshed NetCDF, GeoPackage, and Parquet artifacts plus SHA256SUMS after repairing published `node_order` / boundary-node orientation metadata, ghost coastal outlet `dist_out_dijkstra`, and single-node node-distance conventions.

0.0.3 (March 2026)

- **Routing weights learned from human labels.** Replaced the handcrafted lexicographic 3-tuple `(effective_width, log_facc, pathlen)` with a weighted scalar score trained on 1,967 human-labeled junction decisions: '1.97*log1p(ew) + 0.23*log1p(facc) - 0.23*log1p(slope) + 0.23*log1p(pathlen)
  - 0.29*stream_order'. Two new signals vs prior releases: slope (negative = prefer lower gradient) and stream_order. All routing functions use the same score to prevent divergence. (Weights retrained in 0.0.4; see above.)
- **Caroline's reviewer fixes synced.** 127 C001 lakeflag fixes (NA) and 10 C004 type fixes (NA) from the Streamlit reviewer app.
- **NA PostgreSQL geometry fix.** 38,696 NA reaches had NULL geometry in PostgreSQL — v17b geometry overwrite via dblink silently failed. Rewrote to read from v17b GPKG files on disk with verification gates.
- Fixed `dn_node_id`, `up_node_id`, and `node_order` for 810 flow-corrected reaches where node ordering was stale (based on v17b `dist_out`). 639 reaches now have `dn_node_id != min(node_id)`, reflecting the inverted flow direction.
- Removed redundant `rch_id_up_1..4` / `rch_id_dn_1..4` vector variables from NetCDF export. Only the `[4, N]` matrices `rch_id_up` and `rch_id_dn` are exported, matching v17b format.
- Fixed `swot_orbits` NetCDF type from string back to int64 (matching v17b).
- Fixed 15 integer columns widened from int32 to int64 in NetCDF export (`n_nodes`, `lakeflag`, `n_rch_up`, `stream_order`, etc.). All shared variable types now match v17b exactly.
- Synced `rch_id_up_1..4` / `rch_id_dn_1..4` DB columns from `reach_topology` table (694 were stale after flow corrections).

0.0.2 (March 2026)

- Renamed `is_mainstem_edge` to `is_mainstem`
- Mainstem algorithm refactored: computed per `main_path_id` group (see Section 2.1)
- Fixed `n_rch_up`/`n_rch_down` stale counts at 148 flow-corrected reaches
- Recomputed `facc` at 807 flow-corrected reaches using topological propagation; resolves all F006 junction conservation violations and T003 monotonicity violations at these reaches
- OC reach 51111300061 incomplete split fully reverted to v17b state (434 orphan centerlines, 73 orphan nodes restored)

0.0.1 (March 2026)

- Initial v17c beta release
- Includes `node_order`, `dn_node_id`, `up_node_id` (node ordering within reaches)

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1. Overview

SWORD v17c extends v17b with three additions: computed mainstem topology, SWOT observation statistics, and a rederived flow accumulation (`facc`, 0.0.13). No reaches, nodes, or centerlines were added or removed. v17c contains the same 248,673 reaches, 11.1M nodes, and 66.9M centerline points as v17b across all six regions (NA, SA, EU, AF, AS, OC).

Each region is distributed as a single NetCDF4 file (`{region}_sword_v17c.nc`). The group structure matches v17b (centerlines, nodes, reaches), and the `area_fits` and `discharge_models` subgroups under reaches pass through from v17b unchanged. Reach arrays are ordered stably by `reach_id`. Node arrays are grouped contiguously by `reach_id` and ordered within each reach by `node_order` (downstream to upstream).

Reach coordinate columns (`x`, `y`, `x_min`, `x_max`, `y_min`, `y_max`) match v17b values across all formats (NetCDF, DuckDB, PostgreSQL). Node coordinate columns (`x`, `y`) also match v17b after the 0.0.12 continuity revert.

All new variables use a fill value of -9999 where no observation or computation produced a value.

For a complete variable catalog, see v17c_variable_reference.md.

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2. New Variables

2.1 Mainstem Topology (reaches group)

`is_mainstem` is computed per `main_path_id` group: each group (a mainstem path plus its tributary branches) gets one canonical chain, identified by a greedy walk from the group's shared `best_headwater`. At each junction the algorithm selects the upstream branch with the highest weighted routing score: `2.02*log1p(ew) + 0.17*log1p(facc) - 0.08*log1p(slope) + 0.35*log1p(pathlen) + 0.23*stream_order`. These weights were learned from 1,967 human-labeled junction decisions via logistic regression on pairwise log1p-difference features, using SWOT slope where reliable (else MERIT DEM). The negative slope weight captures the geomorphic pattern that mainstem channels have lower gradients than tributaries. Mainstem reaches within each group are then assigned `rch_id_up_main` / `rch_id_dn_main` from the chain; non-mainstem reaches use the same weighted score. Ghost reaches (type=6) are excluded from mainstem but still participate in routing topology. ~10% of mainstem reaches have no mainstem neighbor at `main_path_id` group boundaries — this is expected by design.

Variable	Type	Units	Description
`dist_out_dijkstra`	float64	meters	Dijkstra shortest-path outlet distance reported at the reach upstream endpoint (outlet = `reach_length`; values retained for ghost reaches)
`hydro_dist_out`	float64	meters	Mainstem outlet distance reported at the reach upstream endpoint via `rch_id_dn_main` chain (outlet = `reach_length`)
`hydro_dist_hw`	float64	meters	Mainstem headwater distance reported at the reach upstream endpoint via `rch_id_up_main` chain (headwater = 0)
`rch_id_up_main`	int64	—	Main upstream neighbor reach_id (mainstem-preferred)
`rch_id_dn_main`	int64	—	Main downstream neighbor reach_id (mainstem-preferred)
`best_headwater`	int64	—	Routing-score-prioritized headwater reach_id for the network component
`best_outlet`	int64	—	Routing-score-prioritized outlet reach_id for the network component
`pathlen_hw`	float64	meters	Cumulative path length from `best_headwater`
`pathlen_out`	float64	meters	Cumulative path length to `best_outlet`
`is_mainstem`	int32	—	1 if reach is on a mainstem path, 0 otherwise
`main_path_id`	int64	—	Unique identifier for each mainstem path group
`subnetwork_id`	int32	—	Connected component ID (Pfafstetter-offset, globally unique; see Section 4)
`dn_node_id`	int64	—	Node ID at the downstream end of the reach (lowest `dist_out`)
`up_node_id`	int64	—	Node ID at the upstream end of the reach (highest `dist_out`)

Eight of these variables also appear at node level. Six are interpolated by node position within the reach: `dist_out`, `hydro_dist_out`, `hydro_dist_hw`, `dist_out_dijkstra`, `pathlen_hw`, and `pathlen_out`. Three are flat copies from the parent reach: `subnetwork_id`, `best_headwater`, and `best_outlet`. Historical flow-correction topology flips were fully reverted in 0.0.9, so the current 0.0.12 database does not retain a flow-corrected reach set requiring reversed `node_id` order.

All six interpolated distance columns use midpoint offsets: `offset = cumsum(node_length) - 0.5 * node_length`. `dist_out`, `hydro_dist_out`, `dist_out_dijkstra`, and `pathlen_hw` use `reach_value - reach_length + offset`; `hydro_dist_hw` and `pathlen_out` use `reach_value + reach_length - offset`. This places each node at the geometric center of its `node_length` segment. On a single-path network (no junctions), node-level `dist_out`, `hydro_dist_out`, and `dist_out_dijkstra` are exactly equal.

`node_order` is a node-level variable (not in the reaches table): 1-based position within a reach, ordered by `dist_out` ascending (1 = downstream end, n = upstream end).

Distance convention note. These are endpoint-reported reach scalars, not traversal-origin claims. `dist_out`, `hydro_dist_out`, and `dist_out_dijkstra` are outlet-distance values reported at the reach upstream endpoint and assign `reach_length` at the outlet. `hydro_dist_hw` is also reported at the reach upstream endpoint, but measures distance from `best_headwater` and assigns 0 at the headwater. See the variable reference for the full convention table.

2.2 SWOT Observation Statistics

Percentile-based summaries computed from available SWOT observations. All percentile, range, and MAD variables share the units of the underlying measurement.

Reaches and nodes:

Variable	Type	Units	Description
`wse_obs_p10`–`wse_obs_p90`	float64	meters	WSE percentiles (10th through 90th, in steps of 10)
`wse_obs_range`	float64	meters	WSE observation range (p90 - p10)
`wse_obs_mad`	float64	meters	WSE median absolute deviation
`width_obs_p10`–`width_obs_p90`	float64	meters	Width percentiles
`width_obs_range`	float64	meters	Width observation range
`width_obs_mad`	float64	meters	Width median absolute deviation
`n_obs`	int32	—	Total SWOT observation count

Reaches only:

Variable	Type	Units	Description
`slope_obs_p10`–`slope_obs_p90`	float64	m/m	Slope percentiles
`slope_obs_range`	float64	m/m	Slope observation range
`slope_obs_mad`	float64	m/m	Slope median absolute deviation
`slope_obs_adj`	float64	m/m	Adjusted slope
`slope_obs_slopeF`	float64	—	Slope F-statistic
`slope_obs_reliable`	int32	—	0 = unreliable, 1 = reliable
`slope_obs_quality`	int32	—	Integer quality category (0–8; see Section 3)
`slope_obs_n`	int64	—	Number of RiverSP node observations used in pass-level slope fits
`slope_obs_n_passes`	int64	—	Number of SWOT passes used
`slope_obs_q`	int64	—	Bitfield quality flag (see Section 3)

2.3 Flow Accumulation (rederived in 0.0.13)

`facc` on every node and reach is rederived from MERIT Hydro upstream area (UPA): section-level D8 matching where the match holds, a probabilistic section matcher where it was withheld, bridges across short gaps, and a

network-constrained fill for the rest (see the 0.0.13 changelog entry). `facc_quality` records the source and tier of every value.

Variable	Type	Group	Description
`facc`	float64	reaches, nodes	Upstream drainage area (km^2), rederived from MERIT Hydro UPA on all nodes and reaches. Reach value = junction-aware reach value (the node maximum on all but 49 reaches).
`facc_quality`	int32 (NetCDF), string (other formats)	reaches, nodes	Provenance `rederived_v1:<source>:<tier>[:review]`; NetCDF code `100*source + 10*tier + review` (Section 3)

A running maximum is applied within sections, so within each reach node `facc` never increases upstream. Values are not summed at junctions and are not raised to an upstream tail where the two independently matched sections disagree, so junction conservation (F006) and downstream monotonicity across junction inflows (T003) are not enforced.

Releases 0.0.1 to 0.0.12 carried a two-stage denoise (`denoise_v3`) on 95,880 reaches and v17b values elsewhere. Its junction rule only ever raised values, so wrong-river samples in v17b were summed into the mainstems; 0.0.13 replaces it. The denoise is described in facc_correction_methodology.md.

2.4 Other New or Updated Variables

Variable	Type	Group	Description
`type`	int32	reaches	Reach classification (1=river, 3=lake_on_river, 4=dam, 5=unreliable, 6=ghost). Not present in v17b NetCDF; added in v17c so NetCDF users can filter by reach type without needing the database.
`dl_grod_id`	int64	reaches	DL-GROD (Deep Learning Global River Obstruction Database; He et al. 2025) dam/obstruction ID
`edit_flag`	string	reaches	Tag for manually edited reaches (e.g., `lake_sandwich`, `harp_lake`)

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3. Flag Encoding Reference

facc_quality

String form (DuckDB, GeoPackage, GeoParquet): `rederived_v1:<source>:<tier>`, with `:review` appended when the row is flagged `junction_inconsistent` or `topology_suspect`. Example: `rederived_v1:soft:B:review`.

NetCDF form (int32): `code = 100*source + 10*tier + review`. Example: `rederived_v1:soft:B:review` = 231.

Source	Code	Meaning
`engine`	1	Section-level D8 match: the node takes the maximum UPA over its own stretch of the matched MERIT route
`soft`	2	Probabilistic section matcher (distance as evidence, D8 connectivity as transition model)

Source	Code	Meaning
`bridge`	3	Held or interpolated from the nearest matched values across a gap
`fill`	4	Network-constrained fill: cleaned v17b profile clipped between trusted upstream and downstream values

Tier	Code	Rule that produced the value
`A`	1	Engine value
`A2`	2	Matcher value from a confident matcher section
`B`	3	Matcher value from a less confident section, or fill bounded on both sides
`C`	4	One-sided bridge hold, or fill with one trusted bound or clipped
`D`	5	Bridge interpolated between two anchors, or width split at a bifurcation
`E`	6	Fill with no trusted bound

Reach tier is the worst node tier in the reach (A < A2 < B < C < D < E); reach source is the most common node source. Tiers record which rule set the value, not a measured accuracy. Review: 0 = not flagged, 1 = flagged.

CF attributes: `flag_values` lists all 48 codes (110 to 461) and `flag_meanings` the matching strings with `:` replaced by `_` (for example `rederived_v1_soft_B_review`). Fill value -9999 = no value.

slope_obs_quality

Value	Meaning
0	reliable
1	small_negative
2	moderate_negative
3	large_negative
4	negative
5	below_ref_uncertainty
6	high_uncertainty
7	noise_high_nobs
8	flat_water_noise

CF attributes: `flag_values = [0,1,2,3,4,5,6,7,8]`, `flag_meanings = "reliable small_negative moderate_negative large_negative negative below_ref_uncertainty high_uncertainty noise_high_nobs flat_water_noise"`.

slope_obs_reliable

Value	Meaning
0	Unreliable
1	Reliable
-9999 (fill)	Not computed (no SWOT slope observations)

slope_obs_q (bitfield)

Bit	Value	Meaning
1	1	Negative slope
2	2	Low number of passes
3	4	High variance

Bit	Value	Meaning
4	8	Extreme value
5	16	Clipped

Flags combine by addition. A value of 0 indicates no quality issues. Example: 5 = negative slope (1) + high variance (4).

is_mainstem

Value	Meaning
0	Not on mainstem
1	On mainstem path

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4. Known Limitations

- **SWOT observation coverage:** SWOT statistics are fill_value (-9999) for reaches and nodes lacking SWOT data.
- **facc (0.0.13):** junction disagreements between independently matched sections (`:review`, 1,406 reaches), the 30,674-reach review list, the width-check gap on lake, tidal, ghost and other reaches, wrong SWORD edges such as Loire → Vilaine, and Amazon reach 62253000041 are described in the 0.0.13 changelog entry. The routing attributes were computed with the 0.0.12 `facc` and were not recomputed.
- **Lake sandwich corrections:** 1,252 reaches reclassified to `lakeflag = 1` where a narrow, shorter-than-neighbor reach sat between lake reaches (tagged `edit_flag = "lake_sandwich"`). ~1,755 similar cases remain (narrow connecting channels, chains).
- **HarP lake corrections:** 7,425 reaches reclassified from `lakeflag = 0` (river) to `lakeflag = 1` (lake) based on HarP v1.1 (Hydrography and River Planform) lake classification data. 200,201 child nodes updated to match. Tagged `edit_flag = "harp_lake"`. Existing tags preserved (comma-delimited when multiple apply).
- **area_fits and discharge_models:** Direct copies from v17b. Not recomputed against v17c facc or SWOT values.
- **`subnetwork_id` vs `network`:** `subnetwork_id` uses Pfafstetter- offset enumeration (globally unique). v17b `network` uses per-region 1-based IDs. Different component counts (v17c finds more via weakly connected components; 19 subnetworks span multiple v17b networks). `network` is retained unchanged from v17b.
- **Flow correction oscillation:** 389 reaches (0.16%) in AF/AS/EU/NA/SA had ambiguous WSE slope signals causing bidirectional flow correction scores. These were reverted to v17b topology.
- **main_path_id consistency:** 3,134 reaches have `main_path_id` values inconsistent with current `(best_headwater, best_outlet)` tuples (V013- V015 lint checks). 80 reaches in NA have `best_headwater` pointing to non-headwater reaches. Requires recomputing `main_path_id` from current headwater/outlet assignments.
- **River naming:** 51.2% of reaches are unnamed (NODATA), ranging from 26% (AF) to 69% (OC). 2.6% of mainstem 1:1 links have local name discontinuities (name changes between adjacent reaches with no junction).
- **Width fill values (A003):** ~1,266 reaches have `width=0` (unmeasured) or `width=-1` (GRWL lake fill). These are v17b fill values, not data errors. Present in both v17b and v17c unchanged. The A003 lint check is downgraded to WARNING for this reason.
- **Node geolocation corrections deferred:** The 0.0.8 and 0.0.10 rederived-node coordinate edits were reverted in 0.0.12 to preserve SWOT D0-D2 time-series continuity. Some node sequence/geolocation anomalies inherited from v17b therefore remain. Treat full geolocation repair as a v18 or separately approved release-envelope change.

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5. Quality Audits

Validation checks performed on the v17c data:

Audit	Finding
Geometry	DuckDB geometries (rebuilt from NetCDF) lack endpoint overlap vertices present in v17b (210,533 reaches affected: 173K +1 point, 37K +2 points). `reach_length` unchanged. Reach coordinate columns (`x`, `y`, `x_min`, `x_max`, `y_min`, `y_max`) copied from v17b to ensure consistency across all formats.
Node coordinate continuity	0.0.12 restores v17b node `x`/`y`, `node_length`, `cl_id_min`, and `cl_id_max` for the 344 previously rederived reaches plus OC reach 51111300061 split-revert residue. Global node coordinate diff vs v17b NetCDF: 0.
n_nodes / reach_length	Internally consistent. Zero N008/G002/G003 violations.
path_freq gaps	v17b had 4,952 connected non-ghost reaches with invalid path_freq (0 or -9999). Resolved in v17c; remaining nodata values are correctly attributed to ghost reaches (type=6).
subnetwork_id	3,027 components across 248,673 reaches verified. Pfafstetter banding correct. Zero cross-region collisions. 19 subnetworks (0.6%) span multiple v17b networks (expected).
Topology integrity	T001 (dist_out_dijkstra monotonicity), T012 (referential integrity), T013 (self-reference), T014 (bidirectional): all pass. T005/T007: zero non-reciprocal edges (151 from incomplete flow correction revert resolved in beta 0.0.1). Historical flow-correction topology flips were fully reverted in 0.0.9; `v17c_flow_corrections` is empty in the current database. Node-level `dist_out` is recomputed for all reaches using the midpoint interpolation convention.
n_rch_up/n_rch_down	148 scalar count mismatches corrected (flow corrections flipped reach_topology but did not recalculate counts). Zero mismatches across all 248,673 reaches.
OC reach split revert	Incomplete `break_reaches()` split of OC reach 51111300061 (434 orphan centerlines, 73 orphan nodes) fully reverted to v17b state.
River name formatting	291 formatting issues corrected (separators, whitespace). Automated checks now enforce ";" separator and alphabetical ordering.
Flow direction	Experimental topology flips were ultimately reverted. The 1,112-flip experiment caused ~30K disconnected reaches and was rolled back; the later retained flow-correction family was also fully reverted in 0.0.9 after the scoring tautology was found. Current v17c-0.0.12 does not retain topology that differs from v17b because of this flow-correction pipeline.
HarP lake corrections	7,425 reaches reclassified lakeflag 0 to 1 from HarP v1.1 data. 200,201 nodes propagated. Tagged `edit_flag = "harp_lake"`.
lakeflag/type consistency	Resolved in 0.0.10. All 248,673 reaches now have consistent `lakeflag` and `type`; `type` is authoritative and diverges from the reach ID last digit on 2,316 reaches (0.9%) due to in-place corrections.

For POM (Pierre-Olivier Malaterre) validation results, see pom_validation_report.md.

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6. File Format

- **Format:** NetCDF4 (one file per region)

- **Naming:** `{region}_sword_v17c.nc` where region is `na`, `sa`, `eu`, `af`, `as`, `oc`
- **Groups:** `centerlines`, `nodes`, `reaches`
  - `reaches/area_fits` and `reaches/discharge_models` subgroups (from v17b)
- **Ordering:** Reach and centerline arrays match v17b ordering. Node arrays are reach-contiguous and sorted by `node_order`.
- **Fill value:** -9999 for all numeric variables (int32, int64, float64)
- **Checksums:** SHA256 hashes listed in `SHA256SUMS_0.0.13.txt`
- **Additional formats:** GeoPackage and GeoParquet exports available (reaches and nodes per region, with geometry), plus global GeoParquet and DuckDB files combining all regions

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7. Methodology Documentation

Document	Description
FACC_FABLE_ASSESSMENT_20261001.md	0.0.13 facc rederivation: method, validation, known defects
facc_correction_methodology.md	Facc denoise used in 0.0.1 to 0.0.12 (replaced in 0.0.13)
pom_requests_summary.md	POM validation check tracker (19 checks, production results)
pom_validation_report.md	POM validation production results
v17c_variable_reference.md	Complete variable catalog for NetCDF export
SWORD_v17b_Technical_Documentation.md	v17b baseline reference